Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

FOR FULL VIDEO SOLUTIONS
SCAN THE QR CODE
youtube.com/@hannahkettlemaths

Hannah Kettle MATHS TUTORIALS

Half Past Paper – Week 27

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
1a	Venn diagrams with set notation	4	U748
1b	Venn diagrams with set notation	2	U748
3	Angles on parallel lines, Angles on a line, Angles in triangles	4	U826,U390, U628
5	Finding the area of trapeziums, Solving direct proportion word problems	5	U265,U721
7a	Using standard form with negative indices	1	U543
7b	Standard form with a calculator	2	U161
9	Finding error intervals for truncated numbers	2	U301
11	Using the product rule for counting	3	U369
13	Finding the circumference of circles, Using Pythagoras' theorem, Area of triangles	3	U604,U385, U945
15	Equations of circles and tangents	1	U567
17	Calculating with ratios and algebra	5	U676
19	Solving quadratic inequalities	5	U133
21	Angles in segments and cyclic quadrilaterals, Geometric proofs with angle facts	4	U251,U471
	Total	41	

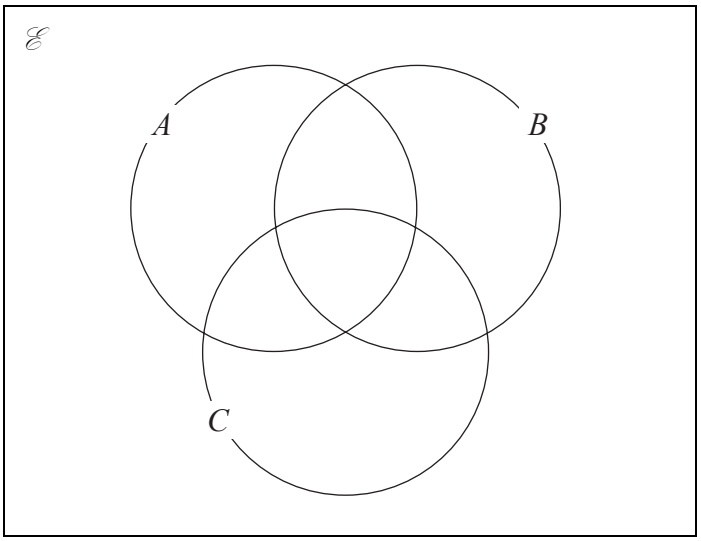
Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 $\mathscr{E}$ = {even numbers between 1 and 25}
A = {2, 8, 10, 14}
B = {6, 8, 20}
C = {8, 18, 20, 22}

(*a*) Complete the Venn diagram for this information.



(4)

A number is chosen at random from $\mathscr{E}$.

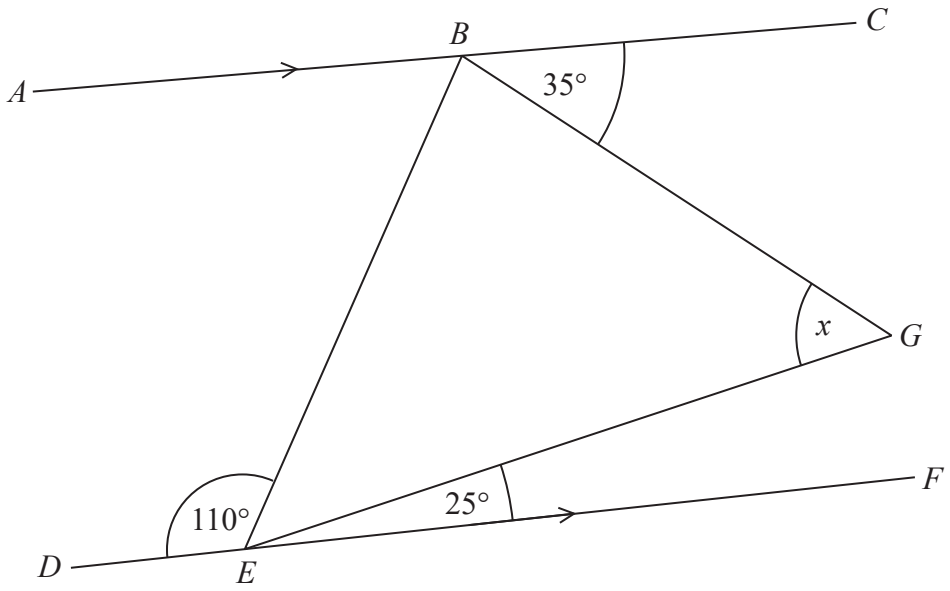
(*b*) Find the probability that the number is a member of $A \cap B$.

.....

(2)

(Total for Question 1 is 6 marks)

3 *BEG* is a triangle.



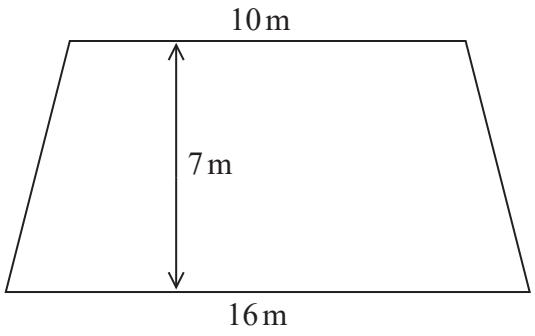
ABC and *DEF* are parallel lines.

Work out the size of angle *x*.
Give a reason for each stage of your working.

.....°

(Total for Question 3 is 4 marks)

5 The diagram shows a floor in the shape of a trapezium.





£16.99

John is going to paint the floor.

Each 5 litre tin of paint costs £16.99
1 litre of paint covers an area of $2m^2$

John has £160 to spend on paint.

Has John got enough money to buy all the paint he needs?
You must show how you get your answer.

(Total for Question 5 is 5 marks)

7 (*a*) Write the number $0.000\,086\,23$ in standard form.

.....

(1)

(*b*) Work out $\dfrac{3.2\times10^{3}+5.1\times10^{-2}}{4.3\times10^{-4}}$

Give your answer in standard form, correct to 3 significant figures.

.....

(2)

(Total for Question 7 is 3 marks)

---

9 Martin truncates the number N to 1 digit.
The result is 7

Write down the error interval for N.

.....

(Total for Question 9 is 2 marks)

---

11 In a restaurant there are

9 starter dishes
15 main dishes
8 dessert dishes

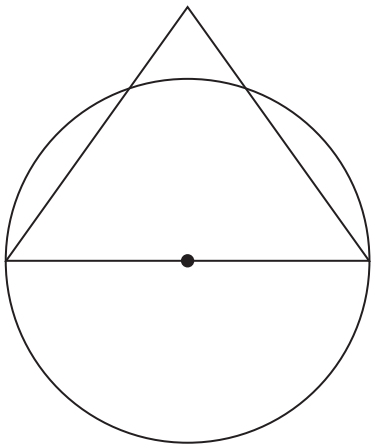
Janet is going to choose one of the following combinations for her meal.

a starter dish and a main dish
or a main dish and a dessert dish
or a starter dish, a main dish and a dessert dish

Show that there are 1335 different ways to choose the meal.

(Total for Question 11 is 3 marks)

13 The diagram shows a circle and an equilateral triangle.



One side of the equilateral triangle is a diameter of the circle.
The circle has a circumference of 44 cm.

Work out the area of the triangle.
Give your answer correct to 3 significant figures.

.....cm^2

(Total for Question 13 is 3 marks)

---

15 The equation of a circle is $x^2 + y^2 = 42.25$

Find the radius of the circle.

.....

(Total for Question 15 is 1 mark)

---

17 p and q are two numbers such that $p > q$

When you subtract 5 from p and subtract 5 from q the answers are in the ratio 5 : 1
When you add 20 to p and add 20 to q the answers are in the ratio 5 : 2

Find the ratio $p : q$
Give your answer in its simplest form.

.....

(Total for Question 17 is 5 marks)

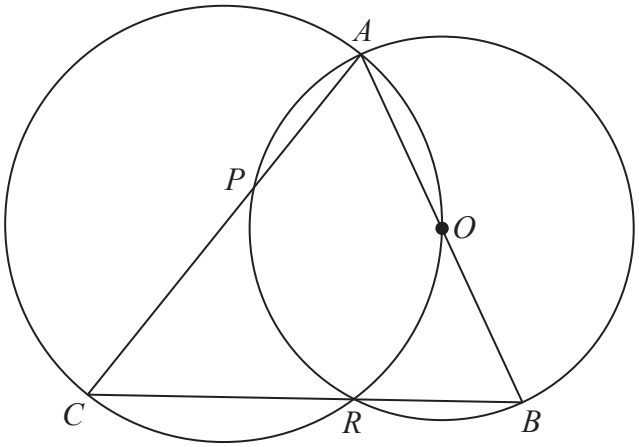
19 Solve $22 < \frac{m^2 + 7}{4} < 32$

Show all your working.

.....

(Total for Question 19 is 5 marks)

21



A, *B*, *R* and *P* are four points on a circle with centre *O*.
A, *O*, *R* and *C* are four points on a different circle.
The two circles intersect at the points *A* and *R*.

CPA, *CRB* and *AOB* are straight lines.

Prove that angle *CAB* = angle *ABC*.

(Total for Question 21 is 4 marks)

TOTAL FOR PAPER IS 41 MARKS